103726

Course Name: Discrete Mathematics

CO1-Understand the notion of mathematical thinking and proofs to solve the problem.
CO2-Apply the basics of discrete probability and number theory to solve the real-world problem.
CO3-Analyze basic discrete structures and algorithms using effectively algebraic techniques.
CO4-Analyze mathematical concepts like sets, reasoning, relational algebra and graph theory to solve optimization problems.
CO5-Analyse the validity of an argument using logical notation.
CO6. Demonstrate the basic structures of proof techniques to write and evaluate the validity of arguments.
CO7. Understand the basic principles of sets, set equalities and operations in sets.
CO8. Apply counting principles to determine probabilities.

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Mid Term Examination, Odd Semester 2025-26
B.Tech. (CS), III Sem, II Year
BCSC 1010 & Discrete Mathematics

Time: 2 Hours **Maximum Marks: 30**

Section – A

Attempt All Questions **3 X 5 = 15 Marks**

No.	Detail of Question	Marks	CO	BL	KL
1	a) A Venn diagram shows two sets, M and N. The number of elements in the region for M only is 12. The number of elements in the region for N only is 8. The number of elements in the intersection is 5. How many total elements are in the union of M and N? b) In a survey of 100 students, 60 like math, 50 like science, and 30 like both. Draw a Venn diagram to represent this information.	3	CO1	U	C
2	If A, B, C are three sets, then show that $A - (B \cap C) = (A - B) \cup (A - C)$	3	CO7	A	P
3	A class of 300 students is taking programming courses.150 students are enrolled in a course on Python.120 students are enrolled in a course on Java.80 students are enrolled in a course on C++.50 students are in both Python and Java.30 students are in both Python and C++.20 students are in both Java and C++.10 students are in all three courses (Python, Java, and C++).How many students are **not** taking any of these three programming courses?	3	CO2	An	C
4	A data center has 100 servers. Each server has a unique IP address within the range of 10.0.0.1 to 10.0.0.200. A network administrator wants to connect these 100 servers to a set of network switches. Each	3	CO3	A	P

	switch can only handle a maximum of 30 connections. What's the minimum number of switches needed to connect all 100 servers?				
5	A student needs to choose 5 courses to take in the upcoming semester. If there are 8 science courses, 6 humanities courses, and 4 mathematics courses available, how many different combinations of 5 courses can the student choose if they must select exactly 2 science courses, 2 humanities courses, and 1 mathematics course?	3	CO8	An	C

Section – B

Attempt All Questions **5 X 3 = 15 Marks**

No.	Detail of Question	Marks	CO	BL	KL
6	a) A number sequence is defined by the rule that each term is n^3+2n. Prove that for any positive integer n, the expression n^3+2n is divisible by 3. b) Suppose that R is the relation on the set of strings of English letters such that aRb if and only if $l(a) = l(b)$, where $l(x)$ is the length of the string x. Is R an equivalence relation?	5	CO6	A	C
7	Imagine you are a data scientist tracking the growth of a new online platform. The number of active users, an on day n follows a specific pattern. Your analysis shows that the change in user growth from one day to the next is directly related to the number of users in the two previous days, with an additional factor related to marketing efforts. The relationship can be modeled by the recurrence relation: $a_{n+2} - 2a_{n+1} + a_n = 2^n$, You know that on the initial day (n=0), there were 2 active users, and on the next day (n=1), there was a dip to only 1 active user. Given this information, what is the exact formula for the number of active users, an, on any given day n? Predict the number of users on the fourth day (n=4).	5	CO3	U	C
8	Consider the set S={2,3,4,6,12,18,24,36} and the relation "divides," denoted by a|b.	5	CO3	A	P

	a) S= {1,2,3,4,6,12,24,36}, is the relation (S,|) a poset (partially ordered set)? Justify your answer by checking the properties of a poset. b) Draw the Hasse diagram for the poset (S,|). c) Based on the Hasse diagram, determine the following: o All maximal elements. o All minimal elements. o The greatest element (if it exists). o The least element (if it exists). d) Find all the upper bounds and lower bounds for the subset {6,12}. Also, find the least upper bound (LUB) and the greatest lower bound (GLB) for this subset. e) Is the poset (S,|) a lattice? Justify your answer using the definition of a lattice				